

Mahima Srivastava, Ph.D.

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Education

Chemical Engineering, Ph.D.

University of Maryland College Park

Aug 2021 – Aug 2026

GPA: 4.0/4.0

Chemical Engineering, B. Tech.

Indian Institute of Technology Kanpur

Jul 2015 – Jun 2019

Professional Summary

Chemical Engineering Ph.D. with expertise in sustainable plastics, biologics formulation, surfactant-protein interactions, and polymer-based material design. Experienced in translating technical briefs into scalable products through cross-functional collaboration, specification development, stability evaluation, and regulatory-conscious documentation. Proven track record in DOE, SOP development, patent strategy, and process optimization from concept to commercialization readiness.

Product Development and Technical Leadership

- Led end-to-end formulation design, optimization, and validation of structured fluids and polymer-based systems.
- Developed product performance standards and rheological specifications to meet defined use requirements.
- Conducted analytical testing and stability assessments to ensure reproducibility and quality compliance.
- Supported scale-up and pilot production through process optimization and cross-functional coordination.
- Contributed to intellectual property strategy and technical documentation for commercialization.

Skills

- **Formulation and Product Development:** Formulations, polymer gels, surfactants, biologics (mAb, ADC), structured fluids, liposomes, emulsions, vesicles, lipid nanoparticles, stability testing, performance benchmarking, ASTM standards, specification development, regulatory documentation
- **Characterization:** Rheology, UV-Vis, DLS/NTA, Zeta Potential, HPLC-MS, DSC, TGA, FTIR, DMA, SEM-EDX, XRD, Tensiometry, Lyophilization, Flow Cytometry
- **Technical Tools:** Python, MATLAB, C++, Java, SQL, Chromeleon CDS, ImageJ, Origin, GraphPad, ChemDraw, Microsoft Office, Technical Writing (SOPs/Reports), ELN systems

U.S. Patent and Invention Disclosure

- **Mahima Srivastava**, Hema Choudhary, Srinivasa R. Raghavan, and Ronald N. Goodman
Plastic products that can be rapidly degraded on-command. U.S. Patent Pub. No. US 2026/0048916 (2026)
- **Mahima Srivastava**, Sai Nikhil Subraveti, Sairam Ganesh, and Srinivasa R. Raghavan
Soft Materials that absorb impact. U.S. Patent Pub. No. US 2024/0353299 (2024)

Work and Collaboration Experience

- **Industrial Research Collaborator**, University of Maryland College Park Oct 2025 – Present
Designed and developed formulation and process workflows for injectable biopolymer-based hemostatic gels, for parenteral delivery via catheter, in collaboration with **Medcura, Inc.**; evaluated formulation composition, rheology, injectability extrusion force through syringe and catheters, shear recovery and stability to establish formulation and process parameters meeting translational performance and quality.
- **Technical Advisor – Patent Commercialization** June 2025 – Present
Collaborated with business development teams to assess market opportunities, and scale-up considerations and supported knowledge transfer of formulation insights to external partners through structured documentation.
- **Programming Analyst**, Citicorp Services Pvt. India. Ltd., Pune, India Aug 2019 – Aug 2021
Built and deployed Docker-containerized Java and Python applications on Kubernetes, creating a scalable and reproducible pipeline for automated data analysis.

Research Experience

Doctoral Research Candidate, University of Maryland

Jan 2022 – Aug 2026

- **Zero-Order Drug Delivery:** Developed controlled-release polymer formulations with defined performance specifications; conducted stability, diffusion, and analytical validation (HPLC, UV-Vis) to ensure reproducibility and quality benchmarks.
- **Stimuli-Responsive Self-Assembly:** Formulated surfactant sol-gel-sol transition system for sensing CO₂, using biocompatible molecules; quantified changes in rheology, self-assembly, and microstructure under controlled gas exposure; established structure–property relationships governing sensitivity, reversibility, and response time.
- **On-Demand Degradable Bioplastics:** Synthesized novel degradable plastic materials meeting ASTM mechanical standards that can degrade within minutes at room temperature; optimized crystallization and processing parameters to enable bench-to-pilot scale transition while maintaining product specifications and performance consistency.
- **Impact-Absorbing Materials:** Developed eco-friendly strong gels capable of absorbing impact forces for protective applications in fragile material packaging; varied design parameters (G' , $\tan \delta$, hysteresis area) to maximize energy dissipation.
- **Imaging Biomarker for PDAC:** Designed avidin-biotin-functionalized liposomal MRI contrast agents carrying biotinylated anti-CA-19-9 antibodies and gadolinium; used SEC to separate antibody-conjugated liposomes from free antibodies.
- **Tissue Mimic:** Performed comprehensive rheological and mechanical characterization of animal tissues in small and large deformation regimes; developed hydrogels with tunable viscoelastic properties to mimic native tissues.
- **Stimuli-Triggered Organogel Degradation:** Developed DBS organogels that can be degraded with a stimulus, with programmable lifetimes (minutes-days); used NMR and MS to explain acid-catalyzed degradation mechanisms.
- **Birefringent Hydrogels for Force-Sensing:** Designed polymer gels exhibiting stress-induced birefringence under cross-polarized light for real-time force sensing; tuned network chemistry and polymer architecture to modulate force sensitivity and optical response thresholds.

Undergraduate Research Assistant, IIT Kanpur

Jul 2018 – Jun 2019

- **Granular Flows Modeling:** Performed simulations for viscoplastic fluid models in COMSOL and discrete element method (DEM) simulations for 2D, steady-state, granular flow dynamics on inclined surfaces, achieving strong correlation between continuum and particle-based models.

Undergraduate Research Assistant, IIT Bombay

Dec 2016 – Jan 2017

- **Identifying Binding Pockets of Sialic-Acid Binding Proteins:** Designed the algorithm to read protein data from RCSB database and identify the probable binding sites of sugar, in Python; verified the classification using SCOP.

Publications

- **Srivastava, M.**, Molkara, R., & Raghavan, S.R.
Ultra-Shear-Thinning Hemostatic Gel Injectable Through a Catheter. *Manuscript ready*.
- **Srivastava, M.**, Choudhary, H., Varisco, D., Hsu, M., Chen, P.Y., Goodman, R. N., & Raghavan, S.R.
Plastic Products that Can be Rapidly Degraded On-Demand. *Under Review*
- **Srivastava, M.**, Nader, M., Varisco, D., & Raghavan, S.R.
Zero-order Drug Delivery Enabled by Polymeric Skin Around a Hydrogel. *Submitted*.
- Colton, A., Halli, R.N., Ma, M.R., Nori, T., Muller, L.K., **Srivastava, M.**, et. al. Geometric Determinants of Sinterless, Low-Temperature-Processed 3D-Nanoprinted Glass. *Microsystems & Nanoengineering*, 11(1), p.145.
- Borden L. K., Nader, M.; Burni, F., Grasso, S., Ortega, I., **Srivastava, M.**, et. al.
Switchable Adhesion of Hydrogels to Plant and Animal Tissues. *Advanced Science*, 2411942.

Leadership & Teaching

Instrumentation Training & Lab Safety Officer

Jan 2022 – Aug 2026

- Trained new members on core laboratory instruments (Rheometer, SEM, DLS, UV-vis, Tensiometer, Lyophilizer).
- Developed SOPs to ensure consistent operation, troubleshooting and data quality.
- Trained lab members on waste management and PPE for 4+ years and maintained compliance with institutional EHS.

Research Mentor

Jan 2022 – Present

- Supervised five undergraduates in our lab group and helped 3 of them to secure financial aid through UMD ASPIRE program.
- Received formal training on being a mentor and mentored a high school student as a part of Center for the Improvement of Mentored Experience in Research (CIMER).

Graduate Teaching Assistant

Aug 2022 – May 2023

- Provided instructional support and led discussions and tutoring sessions for Bionanotechnology and Process Dynamics and Controls.

Conference Presentations

- **Srivastava, M.,** Nader, M., & Raghavan, S.R.
Can hydrogels Mimic the Rheology of Animal Tissues?
ACS Colloids 2026, Newark, DE
- **Srivastava, M.,** Choudhary, H., Varisco, D., Hsu, M., & Raghavan, S.R.
Bioplastics that Can be Degraded On-Command within Minutes.
ACS Colloids 2026, Newark, DE
- **Srivastava, M.,** Nader, M., & Raghavan, S.R.
Can hydrogels Mimic the Rheology of Animal Tissues?
Pacificchem 2025, Honolulu, HI
- **Srivastava, M.,** Choudhary, H., Varisco, D., Hsu, M., & Raghavan, S.R.
Bioplastics that Can be Degraded On-Command within Minutes.
Pacificchem 2025, Honolulu, HI
- **Srivastava, M.,** & Raghavan, S.R.
Designing Gels That Mimic the Rheology of Animal Tissues.
2025 AIChE Annual Meeting, Boston, MA
- **Srivastava, M.,** Choudhary, H., Varisco, D., Hsu, M., & Raghavan, S.R.
On-Demand Degradation of Bioplastics into Soil-Enriching Compost.
2025 AIChE Annual Meeting, Boston, MA
- **Srivastava, M.,** Nader, M., Varisco, D., & Raghavan, S.R.
Are Animal Tissues Elastic or Viscoelastic? *(Invited Poster)*
Annual Burgers Symposium 2025, University of Maryland, MD
- **Srivastava, M.,** Nader, M., Varisco, D., & Raghavan, S.R.
Viscoelasticity of Animal Tissues and Hydrogels. *(Poster)*
Biomedical Engineering Society Meeting (BMES) 2024, Baltimore, MD
- **Srivastava, M.,** & Raghavan, S.R.
How Viscoelastic are Tissues? Insights into Tissue Rheology and on Gels That Can Mimic the Same.
Society of Rheology Meeting 2024, Austin, TX
- **Srivastava, M.,** Choudhary, H., Philip, I. M., & Raghavan, S.R.
Can Plastics be degraded in Minutes? Yes, on-demand. *(Poster)*
ACS Spring Meeting 2024, New Orleans, LA
- **Srivastava, M.,** & Raghavan, S.R.
Using CO₂ to Modulate Self-Assembly and Rheology: Viscosity Change in Surfactant Fluids Induced by CO₂.
International Congress on Rheology 2023, Athens, Greece
- **Srivastava, M.,** Ganesh, S., Subraveti, S. N., Nguyen, D., & Raghavan, S.R.
Designing Gels to Absorb Impact—How a Thin Gel Can Protect an Egg from Breaking.
MRS Spring Meeting 2023, San Francisco, CA
- **Srivastava, M.,** Zhou, C., & Raghavan, S.R.
Using CO₂ to modulate self-assembly: Viscosity increase or decrease in surfactant fluids induced by CO₂.
MRS Spring Meeting 2023, San Francisco, CA

Awards and Honors

- Laboratory Safety Culture Spotlight Award – *University of Maryland College Park 2026*
- MPower Early Scholars Grant – *University of Maryland College Park 2025*
- Kulkarni Foundation Summer Research Fellowship – *University of Maryland College Park 2025*
- “Start a SUD Startup” Challenge Winner – *National Institute on Drug Abuse (NIDA) 2023*
- Goldhaber Travel Award Recipient – *University of Maryland College Park 2023*
- Outstanding Teaching Assistant – *University of Maryland 2023*
- Vibha Gold Medal – *Indian Institute of Technology Kanpur 2019*